

Quantum Electronics & Qubits — Exercise Sheet

6 Solutions

Instructor Notes

Exercise 1: All about Quantum Dots

Part A: Electron Transfer Process

1. Energy Diagram at $V = 0$

At zero bias voltage, the chemical potentials of source and drain are equal:

$$\mu_S = \mu_D = \mu_0$$

The quantum dot (QD) has an energy level that depends on its charge state due to Coulomb charging:

$$E_n = \frac{(ne - Q)^2}{2C_\Sigma}, \quad Q = C_{\text{gate}} V_{\text{gate}}$$

At $V = 0$, the system is symmetric, so the energy levels of the dot align with the Fermi level when $n = 0$ or 1 . The charge quantization is smeared out when:

$$k_B T \gtrsim \frac{e^2}{2C_\Sigma}$$

This corresponds to thermal energy overcoming the Coulomb blockade.

2. Energy Diagram at $V = e/C_\Sigma$ Before Any Transfer

Apply a bias voltage $V = e/C_\Sigma$, so $\mu_S = \mu_0 + eV = \mu_0 + e^2/(C_\Sigma)$, and $\mu_D = \mu_0$.

Energy differences for tunneling processes:

- From source to dot (empty): $\Delta E = E_1 - E_0 = \frac{e^2}{2C_\Sigma}$
- From dot to drain (with one electron): $\Delta E = E_0 - E_1 = -\frac{e^2}{2C_\Sigma}$
- From source to dot (with one electron already): $\Delta E = E_2 - E_1 = \frac{3e^2}{2C_\Sigma}$
- From dot to source (with one electron): $\Delta E = E_0 - E_1 = -\frac{e^2}{2C_\Sigma}$

Only transitions with $\Delta E < 0$ can occur at low temperature $k_B T \ll e^2/(2C_\Sigma)$. Thus, only the process from **source to dot (first electron)** is possible.

3. After One Electron Has Transferred

Now the dot has one electron. The new energy levels are:

$$\begin{aligned} E_0 &= 0 \\ E_1 &= \frac{e^2}{2C_\Sigma} \\ E_2 &= \frac{4e^2}{2C_\Sigma} = \frac{2e^2}{C_\Sigma} \end{aligned}$$

Energy differences:

- From source to dot: $\Delta E = E_2 - E_1 = \frac{3e^2}{2C_\Sigma}$
- From dot to drain: $\Delta E = E_0 - E_1 = -\frac{e^2}{2C_\Sigma}$
- From dot to source: $\Delta E = E_0 - E_1 = -\frac{e^2}{2C_\Sigma}$
- From drain to dot: $\Delta E = E_1 - E_0 = \frac{e^2}{2C_\Sigma}$

Only negative energy differences allow transitions. So the next possible transfer is **from dot to drain**. After this transfer, the dot returns to neutral, and the cycle can repeat.

Part B: Current Through a Quantum Dot

1. Linearized Current and Differential Conductance

Current through the dot:

$$I = \frac{e}{h} \int_{-\infty}^{\infty} d\varepsilon T(\varepsilon) [f_L(\varepsilon) - f_R(\varepsilon)]$$

In linear response ($V \rightarrow 0$), expand:

$$f_L(\varepsilon) - f_R(\varepsilon) \approx -\frac{df_0}{d\varepsilon} \cdot \frac{eV}{k_B T}$$

Then:

$$I \approx \frac{e}{h} \int_{-\infty}^{\infty} d\varepsilon T(\varepsilon) \left(-\frac{df_0}{d\varepsilon} \right) \cdot \frac{eV}{k_B T}$$

So differential conductance:

$$G = \frac{dI}{dV} = \frac{e^2}{hk_B T} \int_{-\infty}^{\infty} d\varepsilon T(\varepsilon) \left(-\frac{df_0}{d\varepsilon} \right)$$

Using Breit-Wigner form:

$$T(\varepsilon) = \frac{\Gamma_L \Gamma_R}{(\varepsilon - \varepsilon_0)^2 + (\Gamma/2)^2}, \quad \Gamma = \Gamma_L + \Gamma_R$$

And derivative of Fermi function:

$$-\frac{df_0}{d\varepsilon} = \frac{1}{k_B T} \cdot \frac{e^{(\varepsilon - \mu)/k_B T}}{(1 + e^{(\varepsilon - \mu)/k_B T})^2}$$

At low temperatures, this becomes a delta peak around $\varepsilon = \mu$. So:

$$G \propto T(\mu)$$

Thus:

$$G(V=0) \propto \frac{\Gamma_L \Gamma_R}{(\mu - \varepsilon_0)^2 + (\Gamma/2)^2}$$

2. Zero-Bias Conductance vs Temperature and Level Position

From above:

$$G(\varepsilon_0) = G_0 \cdot \frac{1}{1 + \left(\frac{\varepsilon_0 - \mu}{\Gamma/2}\right)^2}$$

Temperature broadens the peak via $-df/d\varepsilon$. As $T \uparrow$, peak width increases.

Sketch:

- For $T = 0$: sharp Lorentzian peak centered at $\varepsilon_0 = \mu$
- For increasing T : peak broadens and height decreases

Part C: Thermometry Using a Quantum Dot

1. Width of Conductance Peak and Temperature

In Coulomb blockade regime:

$$\frac{G}{G_\infty} = \frac{\delta/k_B T}{2 \sinh(\delta/k_B T)} \approx \frac{1}{2 \cosh^2(\delta/2.5k_B T)}$$

As T increases, the denominator grows, leading to broader peaks. Therefore, the **width of the conductance peak** is proportional to $k_B T$, which allows us to extract T from the peak width.

2. Full Width at Half Maximum (FWHM)

Let's find δ such that:

$$\frac{G}{G_\infty} = \frac{1}{2} \Rightarrow \cosh^2\left(\frac{\delta}{2.5k_B T}\right) = 2 \Rightarrow \frac{\delta}{2.5k_B T} = \cosh^{-1}(\sqrt{2}) \Rightarrow \delta = 2.5k_B T \cdot \cosh^{-1}(\sqrt{2})$$

Numerically:

$$\cosh^{-1}(\sqrt{2}) \approx 0.881, \quad \text{so FWHM} \approx 2.2k_B T$$

Thus:

$$\text{FWHM} \propto T$$

3. Measurement Protocol

To extract T :

1. Measure $G(V_g)$ at small bias near a Coulomb peak. 2. Fit the peak with a Lorentzian or hyperbolic cosine squared function. 3. Extract FWHM. 4. Use the relation $\text{FWHM} \approx 2.2k_B T$ to infer T .

Assumptions: - System is in the classical Coulomb blockade regime: $E_C \gg \delta E \gg k_B T$ - No level broadening beyond Γ - Equilibrium conditions hold during measurement

4. Systematic Errors

- **Level Broadening:** If Γ is not negligible compared to $k_B T$, it adds to the width $\rightarrow$ overestimates T

- **Noise:** Can distort the peak shape $\rightarrow$ inaccurate fit

- **Nonlinearity:** Deviations from ideal cosh dependence due to higher-order effects or multi-level transport

- **Gate Voltage Calibration:** Inaccurate V_g control affects alignment of ε_0 with μ

Minimization: - Use high-quality devices with well-defined levels - Calibrate gate voltage carefully - Operate in low-noise cryogenic environments - Ensure thermal equilibration before measurement

Exercise 3: Quantum Capacitance of Graphene

1. Concept of Quantum Capacitance

Quantum capacitance arises from the change in electrostatic potential due to changes in the electronic density of states (DOS), rather than geometric separation as in classical capacitance.

It is defined as:

$$C_q = \frac{dQ}{dV} = e^2 \nu(E_F)$$

In graphene, the DOS near the Dirac point is linear in energy:

$$\nu(E) = \frac{|E|}{\pi v_F^2 \hbar^2}$$

Hence:

$$C_q(E_F) = \frac{e^2 |E_F|}{\pi v_F^2 \hbar^2}$$

This makes quantum capacitance especially important in **low-dimensional systems** like graphene where the classical geometric capacitance is large but finite.

2. Experimental Setup

In Xia et al., a **graphene flake** is placed on a Si/SiO substrate and contacted with a gold electrode. An **ionic liquid** covers the exposed part of the graphene and acts as a gate dielectric.

A **three-electrode configuration** is used:

- Graphene as working electrode
- Platinum reference electrode (RE)
- Counter electrode (CE)

The gate voltage V_{TG} controls the carrier density in graphene.

The **ionic liquid** provides a large geometrical capacitance due to the formation of an electric double layer at the interface with graphene.

3. Extraction of Quantum Capacitance

Total measured capacitance C_{total} is the series combination:

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_{\text{geo}}} + \frac{1}{C_q} \Rightarrow C_q = \left(\frac{1}{C_{\text{total}}} - \frac{1}{C_{\text{geo}}} \right)^{-1}$$

Geometrical capacitance per unit area:

$$C_{\text{geo}} = \frac{\varepsilon_0 \varepsilon_r}{t}, \quad \varepsilon_0 = 8.85 \times 10^{-12} \text{ F/m}, \quad \varepsilon_r \approx 7, \quad t \approx 0.3 \text{ nm} \Rightarrow C_{\text{geo}} \approx 200 \mu\text{F}/\text{cm}^2$$

Plotting C_q vs. gate voltage shows a minimum at the Dirac point (zero doping), and increases linearly away from it.

4. Behavior of $C_q(V)$

Since $E_F = eV$, and $C_q \propto |E_F|$, we have:

$$C_q(V) \propto |V|$$

This matches the experimental data shown in Fig. 1(b), where C_q increases linearly with $|V|$, reaching 10 F/cm^2 at $V = \pm 0.3 \text{ V}$.

Discrepancy: - Theory predicts perfect linearity, but experiments show slight deviations near the Dirac point. - This may be due to: - Residual disorder or inhomogeneity in the sample - Finite screening effects - Nonlinearities in the gate voltage calibration

5. Challenges in Measuring Quantum Capacitance with Ionic Liquid

- **Electrochemical Stability:** Ionic liquids may decompose under high voltages
- **Screening Effects:** Mobile ions can screen the electric field $\rightarrow$ reduce apparent C_q
- **Parasitic Capacitances:** Wiring and setup contribute to total capacitance
- **Thermal Fluctuations:** At higher temperatures, C_q broadens
- **Sample Quality:** Defects and impurities alter the DOS

These challenges can affect the accuracy of the extracted C_q , particularly near the Dirac point.